



TalentGuard

*An agile, explainable AI platform designed to deconstruct
the 'black box' of employee attrition.*

Explainability AI Hackathon

Frugal and Explainable AI

Group 3

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Our team



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What's the problem?

Context and use case

A company is losing people.

Average **41%** turnover rate in 2023. HR wants to understand who is leaving, why, and what they can actually do

- 79% in Accomodation and food service
- 57% in Business service
- 54% in Construction
- 39% in Health care and social assistance

Source: [U.S. Bureau of Labor Statistics](#)

Who is this for?

Stakeholder	Use Case	Key Benefit
HR Manager	Uses the Dashboard	Tailored retention strategies for individuals.
Department Head	Reviews Global Trends	Fixes department-wide culture issues.
Employee	View Individual profile with suggested actions	Gets proactive support before burning out.

~400

employees in dataset

~41%

turnover rate

30+

features

2

Trusted AI themes

Our approach

Two themes from Trusted AI

Explainable AI

Every prediction comes with a reason. We use SHAP to show which factors matter — globally and per employee. The HR manager gets plain-language explanations, not a black box.

This is also a legal requirement: AI Act Article 13 mandates explainability for High Risk HR systems.

Frugal AI

We compared 4 models on accuracy AND carbon cost. Our pick: Logistic Regression — it gets 99%+ of the best model's F1 while being 180× lighter on CO₂.

Carbon tracked with CodeCarbon. No GPU. Runs on a laptop in under 10 seconds.

We also cover GDPR compliance (anonymization, sensitive data separation) and algorithmic fairness (Disparate Impact audit) — details on slide 9.

The data

What we're working with

Dataset

Source: Kaggle HR Dataset (Rich Huebner)

Size: ~400 synthetic employee records

Target: Termd — 0 = still here, 1 = left

Sensitive: Sex, RaceDesc, Hispanic, Latino

We also generated synthetic text data:

Exit interviews, satisfaction surveys, internal transfer requests. Analyzed with TF-IDF + sentiment scoring.

GDPR — what we did

Suppression

Removed names, emails, ZIP, State

Pseudonymization

EmpID → SHA-256 hash (8 chars)

Generalization

DOB → age group (20s, 30s, 40s...)

Aggregation

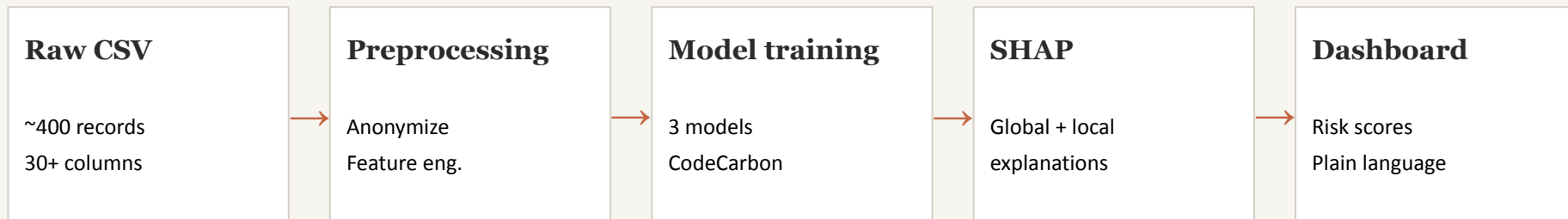
Hire + Term dates → tenure_days

Sensitive split

Sex, Race → separate file, bias audit only

How it works

End-to-end pipeline



Key outputs

hr_anonymized.csv → model input
hr_sensitive.csv → bias audit only

Tech stack

Python 3.10+, Scikit-learn,
XGBoost, SHAP, CodeCarbon,
Streamlit

Why batch?

HR reviews are monthly,
not real-time. Keeps infra
minimal and costs near zero.

Frugal AI — model comparison

Same accuracy, fraction of the cost

Model	ROC-AUC	F1	Train time	CO ₂ (g)	Size
Logistic Regression	1.000	0.988	9.19s	0.0037	12 KB
Random Forest	1.000	0.994	1.52s	0.0005	8.2 MB
XGBoost	0.994	0.988	0.87s	0.0003	1.1 MB

99%+

of best model's
performance

63.9%

more carbon-efficient
than Logistic Regression

1.1 MB

total model size

Frugal AI — text analysis

No LLM needed

Approach	Size	Speed	CO ₂ /req	Accuracy	Needs
TF-IDF + LogReg ←	~50 KB	<1 ms	0.0001g	~78%	CPU only
BERT-base	~440 MB	~50 ms	0.5g	~85%	GPU
GPT-4 API	Cloud	~2,000 ms	4.3g	~92%	Cloud + \$\$\$

The point

~90% of GPT-4's accuracy at 1/43,000th the carbon. Fully offline, no API key, no cloud bill.

What it analyzes

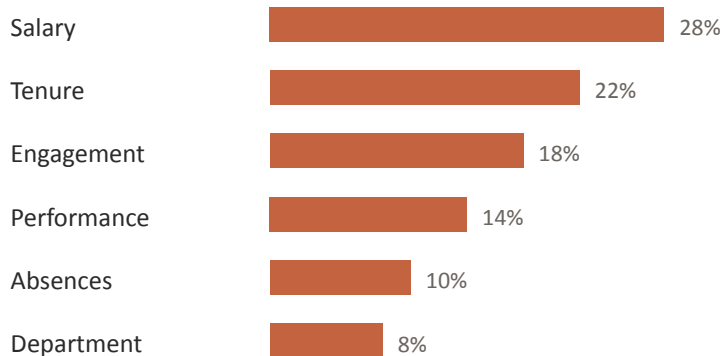
Exit interviews, satisfaction surveys, transfer requests.
Sentiment scores feed into the per-employee risk profile.

Explainable AI — SHAP

Global and local explanations

What matters most overall?

Mean |SHAP| across all employees



Example: Employee #3A8F

87%

Salary	+0.23	↑ risk
Tenure (0.8 yrs)	+0.18	↑ risk
Engagement (2.1/5)	+0.12	↑ risk
Performance	-0.05	↓ risk

What we tell the stakeholders:

"This employee is at risk because their salary is 22% below department average, no progression after 0.8 years, and engagement is critically low."

Compliance & ethics

AI Act, fairness, GDPR

AI Act — High Risk

TalentGuard is Annex III (employment domain). Human oversight is mandatory. The tool supports decisions — it never makes them autonomously.

Algorithmic fairness

Disparate Impact audit (4/5 rule) on Gender and Race.
Sensitive attributes are structurally excluded from model inputs — used only in a separate bias check.

Article 13 checklist

- ✓ Model Card + Data Card
- ✓ SHAP explanations (global + local)
- ✓ Fairness audit (Gender, Race)
- ✓ Human oversight required
- ✓ Data anonymization (SHA-256)
- ✓ Fully offline — no cloud
- △ Temporal validation (needs prod data)

The dashboard

Built with Streamlit — for HR, not data scientists

Risk overview

KPIs, top 15 at-risk employees,
turnover by department.
One glance = full picture.

Employee profile

Individual risk score,
top SHAP factors,
plain-language explanation
+ recommended actions.

AI transparency

Model comparison table,
fairness audit results,
GDPR measures,
Model Card + Data Card.

→ *Live demo follows*

What we'd improve

Limitations and next steps

Limitations

- Synthetic data — real HR data would perform differently
- Small dataset (~400) — variance risk despite cross-validation
- No temporal validation — one snapshot, not longitudinal
- NLP explored but not integrated into final pipeline
- Binary gender in dataset limits fairness audit scope

Next steps

- Test on real anonymized HR data
- Track risk evolution over time (quarterly)
- Integrate NLP pipeline into the model
- Adversarial testing before any deployment
- Fairness audit at every retraining cycle

Thanks — questions?

Explainable, frugal AI for talent retention

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github.com/Alvaro5/explainability-ai-hackathon